

Heparin, LMWH, Fondaparinux, Direct thrombin inhibitors, Xa inhibitors

1. Beaver dam: fibrin clot
2. Throm-beaver: thrombin
3. Throm-beaver preparing stick: thrombin transforms fibrinogen into fibrin, these fibrin monomers will crosslink with calcium, a phospholipid surface and factor XIII to make a strong mesh
4. Throm-beaver II shaped teeth: factor II (thrombin)
5. FoX: factor X and Xa catalizes the conversion of pro-thrombin into thrombin (factor 2 into 2a)
6. FoX waking up throm-beaver: factor Xa converts prothrombin into thrombin.
7. **Heppy hunting: heparin**
8. Heppy hunting father: unfractionated heparin
9. Trap with III shaped bars: unfractionated heparin binds with antithrombin III
10. Trapped throm-beaver and foX: the unfractionated heparin-antithrombin III complex irreversible inactivates thrombin and Factor Xa
11. Birdwatching father: monitor APTT (activated partial thromboplastin time) to assess unfractionated heparin levels
12. Woodpecker inside tree trunk: PTT measures the function of the intrinsic pathway of the coagulation cascade
13. Heparin is not a clot buster, it will prevent the fibrin clots from forming
14. Hunting at the iliofemorial river: heparin can be used for deep vein thrombosis prophylaxis
15. Beaver dam ion the iliofemoral river: heparin can be used for acute treatment of deep vein thrombosis
16. Lung shaped tree: pulmonary arterial tree
17. Birds nest on ischemic branch: heparin can be used for prophylaxis and acute treatment of pulmonary embolism (PE)
18. Branched distal is ischemic and leafless
19. Continuous heparin drip can help reduce the development of the PE
20. Broken heart strings: Heparin is used in the setting of an acute MI, heparin is used to prevent clot extension and also formation
21. Ivy: administration IV heparin in the setting of acute DVT, PE, and MI
22. **Adverse effects of heparin**
23. Bleeding, need to keep close monitoring PTT
24. Shooting four clay plates: Heparin induced thrombocytopenia, (ex. in a 50 y/o pt receiving heparin prophylaxis for a few days then gets a swollen foot) This occurs when antibodies are made against heparin complex to platelet factor 4
25. Broken plates: heparin can cause thrombocytopenia
26. Throm-beaver dam around broken plates: HIT results in paradoxical thrombosis in the setting of thrombocytopenia
27. Depleted mineral mine: heparin can cause hyperaldosteronism (a mineralocorticoid) even in low doses heparin causes this
28. Big K: heparin induced hypoaldosteronism (type 4 RTA) causes hyperkalemia
29. Porous termite damage: heparin can cause osteoporosis
30. Protected area deterring the hunter: protamine sulfate reverses the anticoagulant effect of the unfractionated heparin (less effective foe LMWH and fondaparinux)
31. Protamine sulfate is a positively charged peptide that binds to unfractionated heparin (negatively charged)
32. **Heppy hunter daughter: low molecular weight heparin (LMWH)**
33. Trap with III: LMWH binds antithrombin III: Similar MOA as Unfractionated heparin,
34. FoX in small trap: the LMWH-antithrombin complex inhibits factor Xa with less of an effect on thrombin
35. Protected area deterring the hunter: protamine sulfate reverses the anticoagulant effect of unfractionated heparin (less effective against LMWH and fondaparinux)
36. Long tapering flag: LMWH has a prolonged half-life
37. Does not require PTT monitoring



38. Decay flag breaking a kidney shaped rock: LMWH is eliminated renally and can stay in the system if there is renal insufficiency
39. Heppy pregnant hunter: heparin is safe during pregnancy. “keep that baby heppy with heparin”
40. Intact clay pigeons: LMWH is less likely to induce HIT
41. ALL TYPES OF HEPARIN CAN CAUSE HIT, Just less likely
42. **Fido with a pair of FoXes: fondaparinux**
43. Protected area deterring the hunter: protamine sulfate reverses the anticoagulant effect of the unfractionated heparin (less effective foe LMWH and fondaparinux)
44. Fido with two cages: Fondaparinux binds antithrombin III with higher specificity that LMWH
45. FoX in small trap: the fondaparinux-antithrombin complex inhibits factor Xa with less of an effect on thrombin
46. Fido with no broken plates: Lowest risk of HIT
47. **NO intruding: bivaliRUDIN is a direct thrombin inhibitor**
48. **Big GATOR: arGATROban and dabigatran are direct thrombin inhibitors**
49. Intruding gator directly eating the throm-beaver: use direct thrombin inhibitors (argatroban, dabigatran, bivalirudin) in HIT
50. **Banned foXes: direct factor Xa inhibitors rivaroxaban and apiXaBAN**
51. Directly grabbing foX: factor Xa inhibitors bind directly
52. Open mouth: factor Xa inhibitors are oral medications, Afib
53. Irregularly irregular tv signal: direct Xa inhibitors are used for long term anticoagulation in atrial fibrillation

